# TurnOne: The Cost of Convention in Low-Rank Games

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## Abstract

Expert Pokémon VGC strategies look nothing like Nash equilibria (total variation 0.99), yet expert-vs-expert outcomes match Nash payoffs within 0.02. This is not expert skill: random strategies achieve the same result. SVD of the $\sim$200-action payoff matrices reveals effective rank 3, with 93% of strategic variation in the payoff null space. Switching from behavioral cloning to offline RL (Conservative Q-Learning) confirms that game structure, not learning objective, is the binding constraint: CQL reshapes 60% of the distribution but gains only 17% in exploitability. The one exception is terastallization—an irreversible resource where convention accounts for 30% of total exploitability.

## 1. Introduction

Here is a puzzle. In competitive Pokémon VGC, expert strategies are maximally different from Nash equilibria (TV $= 0.99$) yet produce identical payoffs against each other (gap $= -0.02$). It is not because experts are clever: random strategies achieve the same result. Something about the game itself makes convention free.

We study this in the VGC doubles format, where both players simultaneously choose actions for two active Pokémon on turn 1. The action space is combinatorial ($\sim$200–400 valid actions per side), the game is zero-sum and simultaneous, and 154,718 recorded expert battles provide a large behavioral dataset. Our goal is not to build a stronger agent but to explain *why* conventional play succeeds—a structural question that agent-building alone cannot answer.

We formalize turn-1 play as an offline contextual bandit—context is the game state, action is one of $\sim$200 valid turn-1 combinations, reward comes from a learned dynamics model—and ask three questions:

- **Convention $\neq$ strategy.** How far is expert play from optimal? Very far: BC is exploitable by 1.41 reward units and TV $=$ 0.99 from Nash—yet cross-play outcomes match Nash payoffs within 0.02 (§3).
- **Convention is free.** Why doesn't this matter? Payoff matrices have effective rank 3 out of $\sim$122 actions, so 93% of strategic variation is payoff-irrelevant (§4). The exception is terastallization, an irreversible resource that accounts for 30% of exploitability (§5).
- **Can reward optimization escape?** Switching from imitation to offline RL (CQL) reshapes 60% of the distribution but gains only 17% in exploitability—94% of the shift lands in the payoff null space (§6).

### 1.1. Related Work

**Minimax in sports.** Studies of penalty kicks (Chiappori et al., 2002; Palacios-Huerta, 2003) and tennis serves (Walker & Wooders, 2001; Anderson et al., 2024) find near-minimax play in 2–3 choice domains. Our setting has $\sim$200 actions and requires a learned payoff model.

**Price of anarchy and focal points.** The price of anarchy (Koutsoupias & Papadimitriou, 1999; Roughgarden & Tardos, 2002) measures the cost of equilibrium play relative to social optimum; we measure the cost of *conventional* play relative to equilibrium. Schelling (1960) argued that players coordinate on salient defaults (focal points); our finding that expert conventions are payoff-equivalent to Nash is a quantified realization of this insight.

**Poker AI.** Superhuman poker agents (Bowling et al., 2015; Brown & Sandholm, 2019) demonstrate that human play is exploitable but focus on building agents rather than characterizing the *structure* of human suboptimality.

**EGTA.** Our approach follows the EGTA framework (Wellman, 2006), substituting a learned dynamics model for the traditional simulator oracle.

**Pokémon AI.** VGC-Bench (Angliss et al., 2025) evaluates LLM agents; Metamon (Grigsby et al., 2025) applies offline RL to singles. Both train agents; we measure existing human play.

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## 2. Approach

We instantiate the contextual bandit as follows. Each turn-1 decision is a context $s$ (team composition, leads, field), an action $a$ from $\sim$200 valid combinations, and a reward $r(s, a, a')$ from a learned dynamics model. The horizon-1 structure means $Q(s, a) = \mathbb{E}[r \mid s, a]$ with no bootstrapping—CQL reduces to conservative reward regression. We compare three approaches: behavioral cloning (BC), which imitates expert play and ignores reward; CQL, which optimizes reward while staying near the data distribution; and Nash equilibria, computed via LP on the same reward model's payoff matrix.

### 2.1. Data

We use 154,718 rated battles from Pokémon Showdown (Gen 9 VGC Regulation G, 1500+ Elo), yielding 309,436 directed turn-1 examples (Brennan-Almaraz, 2026). We exclude battles with voluntary switches ($\sim$27%). Each example encodes full team composition (6 Pokémon with moves, abilities, items), lead assignments, and field state. Actions are encoded as 16 slots per Pokémon (4 moves $\times$ 4 targets) with a 3-way tera flag.

### 2.2. Behavioral Cloning

Our BC model uses a Transformer encoder (4 layers, 4 heads, $d_{\text{model}} = 128$) operating on 12 Pokémon tokens plus 1 field token. Two heads decode independently: a 16-way action head per active Pokémon and a 3-way tera head, with $P(a_1, a_2, \text{tera}) = P(a_1) \cdot P(a_2) \cdot P(\text{tera})$. The model achieves 44.8% top-1 accuracy (3$\times$ uniform) and 79.6% top-3 accuracy with 1.32M parameters. We validate the independence assumption in Section 7: mutual information between leads' actions is 0.018 bits.

### 2.3. Dynamics Model

The dynamics model shares the encoder architecture, adds action embeddings fused with the encoder output, and predicts per-Pokémon HP changes, KO probabilities, and post-turn field state. This defines the bandit's reward function $r(s, a, a')$. It achieves HP MAE of 13.3/100 and KO AUC of 0.91.

**Signal-to-noise validation.** Since all game-theoretic quantities depend on learned dynamics, we validate that signal exceeds model noise. Injecting Gaussian noise at the dynamics model's reward-space error scale (MAE $= 1.19$) yields a noise-floor exploitability of 0.41—a signal-to-noise ratio of 3.4:1 against our measured exploitability of 1.41.

*Table 1.* Payoff landscape for conventional vs. optimized play (500 matchups, mean values).

| Scenario | Payoff (P1 perspective) |
|---|---|
| BC worst-case | $-1.19$ |
| Nash value $V^*$ | $+0.22$ |
| BC-vs-BC | $+0.20$ |
| Best-response to BC | $+1.63$ |
| Exploitability | **1.41** (vs. optimizer) |
| Cross-play gap | **$-$0.02** (vs. convention) |

### 2.4. Game-Theoretic Framework

For each matchup, we enumerate valid joint actions per side, evaluate each pair with the dynamics model, and construct a payoff matrix $R \in \mathbb{R}^{m \times n}$ $(m, n \sim 200)$—the payoff matrix of the contextual bandit. Payoffs combine HP advantage, KO advantage (weight 3.0), and field advantage (weight 0.5), yielding a zero-sum game (Sahinguyen, 2020; OpenAI et al., 2019; Ishii et al., 2018). The weights are heuristic but our findings are invariant to the choice (Section 7). Nash equilibria are computed via LP: mixed strategies $p^*, q^*$ and game value $V^* = {p^*}^\top R\, q^*$.

**Exploitability** measures how much a player loses by deviating from Nash. For player 1 with strategy $p$, exploitability is $V^* - \min_j (p^\top R)_j$: the gap between the Nash value and the worst-case payoff against a best-responding opponent. We report the average over both roles.

## 3. Experts Are Exploitable

We evaluate on 500 randomly sampled test matchups, reporting 95% bootstrap confidence intervals throughout.

**Experts are exploitable.** Mean exploitability is 1.41 [1.36, 1.46] reward units—roughly half a KO advantage per turn. Table 1 shows the payoff landscape: against a best-responder, BC achieves only $-1.19$, while a best-responder to BC achieves $+1.63$.

**Experts are maximally different from Nash.** Total variation distance between BC and Nash strategies is 0.99. Nash equilibria concentrate on just 2.7 actions on average, while BC spreads mass across $\sim$95 actions. Only 1.3% of BC's probability mass falls on Nash-support actions (median 0.19%). By every measure, expert play is nothing like Nash.

**Yet outcomes are identical.** Despite individual exploitability of 1.41 and TV of 0.99, expert-vs-expert payoffs match Nash: the gap $p_{\text{BC}}^\top R\, q_{\text{BC}} - V^*$ is $-0.02$ $[-0.07, 0.04]$, statistically indistinguishable from zero. This is the puzzle we explain next.

*Table 2.* Cross-play gap $p^\top R\, q - V^*$ for different strategy types (500 matchups, 1000 random draws each).

| Strategy type | Mean gap | Mean $\lvert$gap$\rvert$ |
|---|---|---|
| BC (expert) | $-0.02$ | 0.45 |
| Uniform | $-0.02$ | 0.31 |
| Shuffled BC | $-0.02$ | 0.36 |
| Random (Dirichlet) | $-0.02$ | 0.32 |

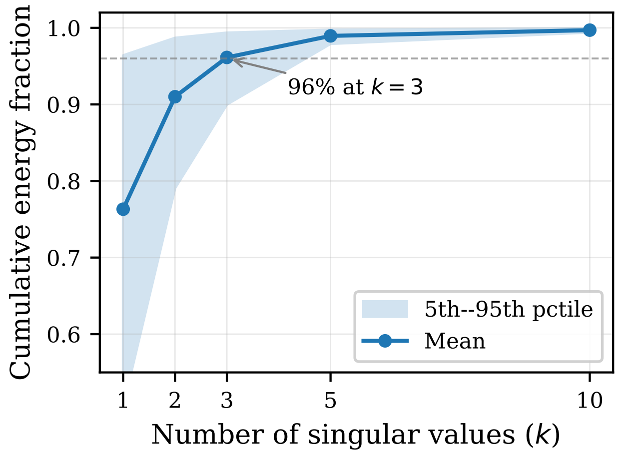


*Figure 1.* Cumulative energy fraction vs. number of singular values, averaged over 500 matchups (5th–95th percentile envelope). The top 3 singular values capture 96% of spectral energy, revealing that VGC turn-1 is a 3-dimensional game in a $\sim$122-dimensional action space.

# 4. Why Convention Is Free

## 4.1. It's the Game, Not the Experts

If the near-zero gap were a property of expert skill, then non-expert strategies should produce larger gaps. They do not. We test three counterfactuals: *uniform* strategies ($1/n$ over all actions), *shuffled* BC (permuted action labels), and *random* strategies (Dirichlet draws). Table 2 presents the results.

All strategy types achieve gaps indistinguishable from zero. The near-zero BC-vs-BC gap is not evidence of expert near-optimality—it is a structural property of the game. Any mixed strategy pair produces near-Nash payoffs because most strategic variation is payoff-irrelevant.

## 4.2. The Game Is 3-Dimensional

Why does the game have this property? SVD of each matchup's payoff matrix reveals effective rank 2.9 [2.8, 2.9] at 95% energy out of $\sim$122 nominal actions. The top singular value captures 76% of the Frobenius norm; the top 3 capture 96% (Figure 1).

*Table 3.* Deviation energy in top-$k$ SVD dimensions (500 matchups, 95% bootstrap CI).

| $k$ | Energy in top-$k$ | In null space |
|---|---|---|
| 1 | 0.022 [0.021, 0.024] | 0.978 |
| 3 | 0.071 [0.068, 0.075] | **0.929** |
| 5 | 0.101 [0.098, 0.105] | 0.899 |
| 10 | 0.151 [0.146, 0.156] | 0.849 |

*Table 4.* Game properties under rank-$k$ approximation (500 matchups).

| $k$ | Exploit ratio | $\lvert\Delta V^*\rvert$ | TV(Nash$_k$, Nash) |
|---|---|---|---|
| 1 | 0.756 | 0.444 | 0.850 |
| 3 | **0.972** | 0.144 | 0.523 |
| 5 | 0.993 | 0.053 | 0.374 |
| 10 | 1.000 | 0.027 | 0.260 |

**Projection test.** We project BC–Nash deviation vectors onto $R$'s singular subspaces. For player 1, $\|U_k^\top \delta_p\|^2/\|\delta_p\|^2$ measures the fraction of deviation energy in the top-$k$ column space (analogously for player 2). Only 7.1% [6.8%, 7.5%] falls in the top-3 dimensions—**93% of deviation energy lies in the payoff null space** (Table 3).

## 4.3. Interaction Decomposition

The low rank explains why the bilinear interaction term vanishes. Decomposing the payoff gap:

$$\underbrace{p_{\text{BC}}^\top R\, q_{\text{BC}}}_{\text{BC-vs-BC}} - V^* = \Delta_1 + \Delta_2 + \underbrace{(p_{\text{BC}} - p^*)^\top R\,(q_{\text{BC}} - q^*)}_{\text{interaction}} \tag{1}$$

where $\Delta_1 = p^{*\top} R\,(q_{\text{BC}} - q^*)$ and $\Delta_2 = (p_{\text{BC}} - p^*)^\top R\, q^*$ are individual deviation costs. The minimax theorem guarantees $\Delta_1 \geq 0$ and $\Delta_2 \leq 0$, but their near-cancellation in magnitude is empirical, not a theorem: over 500 matchups, $\Delta_1 = +1.03$ and $\Delta_2 = -1.02$, with per-matchup std of 0.57. The interaction term is $-0.03$—small because the deviations $\delta_p, \delta_q$ are nearly orthogonal to $R$'s column and row spaces.

**Rank-reduction validation.** If the game is truly 3-dimensional, rank-$k$ approximations should preserve game-theoretic properties. Table 4 confirms: rank-3 preserves 97.2% of exploitability.

**Spectral bound.** Standard spectral arguments formalize why low rank implies small gaps. Let $R = R_k + E$ where $R_k$ is the best rank-$k$ approximation and $\|E\|_2 = \sigma_{k+1}$ by the Eckart–Young theorem (Eckart & Young, 1936). For any strategy pair $(p, q)$: $|p^\top R\, q - V^*| \leq |p^\top R_k\, q - V_k^*| + 2\sigma_{k+1}$, where $V_k^*$ is the Nash value of $R_k$. Both the perturbation term and the Nash-value shift are bounded by $\sigma_{k+1}$ via $|p^\top E\, q| \leq \|E\|_2 \leq \sigma_{k+1}$ for probability vectors. For VGC with $k = 3$ and $\sigma_4/\sigma_1 \approx 0.04$, the empirical gap of 0.02 is

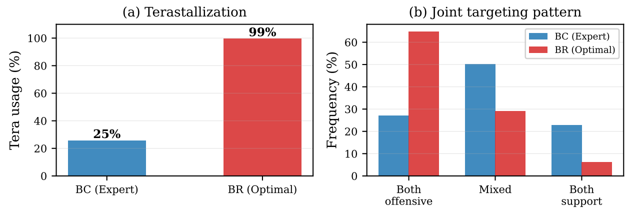


*Figure 2.* Behavioral differences between conventional (BC) and optimal (BR) play. **(a)** Best responses terastallize almost universally (99%) while experts conserve tera for later turns (25%). **(b)** Optimal play is systematically more aggressive, shifting the joint pattern from balanced to heavily offensive.

well within this bound.

**Robustness to model capacity.** Low rank could be an artifact of the dynamics model's bottleneck. We test across three action-embedding capacities ($d = 32, 64, 128$). Effective rank increases only from 2.9 to 4.3, and the highest-capacity model has worse validation loss (526 vs. 445), suggesting extra dimensions reflect overfitting. The two well-fit models agree on effective rank within $\pm 1$ for 93% of matchups. Payoff-weighted TV is stable at 0.019 across all capacities. Low rank is a property of VGC, not a model artifact.

## 5. Where Convention Fails: Terastallization

There is one dimension where convention is costly. Terastallization is an irreversible resource in VGC: once used, it is consumed for the entire game. Experts default to conserving it—BC uses tera on turn 1 in only 25% of matchups. Optimal play uses it 99% of the time.

**Magnitude.** Restricting the best response to BC's tera class (same tera/no-tera choice) reduces exploitability by 0.41 reward units—**30% of total exploitability** [29%, 32%]. This is the single largest behavioral lever.

**Aggression.** Beyond tera, best responses target opponents 79% of the time vs. 52% for BC. The joint action pattern shifts from balanced (27% both-offensive, 50% mixed, 23% both-support under BC) to heavily offensive (65% both-offensive, 29% mixed, 6% both-support under BR). Figure 2 visualizes both contrasts.

**Case study.** A Calyrex-Ice / Ogerpon-Hearthflame lead facing Kyurem-White / Maushold makes the pattern concrete. BC plays Trick Room + Follow Me with 49% probability—a purely defensive setup targeting self—for a payoff of $-0.75$. Nash concentrates 87% on High Horsepower + Horn Leech, both targeting the opponent's Maushold slot, with tera, for a payoff of $+2.75$. The best response copies this aggressive targeting (payoff $+2.76$ against BC), yielding a swing of 3.51—the largest in our sample. Convention learned a defensive posture (Trick Room to reverse speed, Follow Me to redirect attacks) when the game demands immediate aggression. Both tera and target selection contribute: BC never terastallizes in its top action, while Nash always does; and BC targets self in every top action, while Nash targets the opponent exclusively.

*Table 5.* BC vs. CQL vs. Nash (500 matchups, 95% CI).

| Metric | BC | CQL | Nash |
|---|---|---|---|
| Exploitability | 1.40 [1.36, 1.46] | 1.15 [1.08, 1.23] | 0.00 |
| TV($\cdot$, Nash) | 0.99 | 0.98 | 0.00 |
| Worst-case value | $-1.23$ | $-0.99$ | $+0.16$ |

**Domain interpretation.** Experts rationally conserve tera for later turns when they might have more information; our turn-1-only game theory says use it now. This gap between myopic and far-sighted optimality explains the 30% contribution and suggests it would shrink in a multi-turn analysis. A regression of exploitability on structural features yields $R^2 = 0.08$: the *direction* of exploitation is systematic (tera + aggression), but the *magnitude* is matchup-specific.

## 6. Can Reward Optimization Escape Convention?

Every result so far uses imitation learning. Perhaps convention persists only because BC's objective never incentivizes escaping it. We test this by replacing imitation with explicit reward optimization: Conservative Q-Learning (CQL) (Kumar et al., 2020) on the same dataset, using a horizon-1 factored Q-function $Q(s, a) = Q_a(s, \text{slot}_a) + Q_b(s, \text{slot}_b) + Q_{\text{tera}}(s, \text{tera})$ with the same encoder architecture as BC. The factored decomposition is justified by the near-zero mutual information between leads (0.018 bits). CQL's conservative penalty discourages out-of-distribution actions, making it a principled offline RL baseline: it can improve on expert behavior where the reward signal supports it, while staying grounded in the data distribution.

CQL reliably reduces exploitability (411 of 500 matchups, $p < 10^{-51}$), narrowing the gap by 17% overall (1.40 $\rightarrow$ 1.15) and improving worst-case value by 20%. But it does not close the gap: both policies remain TV $= 0.98$ from Nash. The improvement is not uniform—CQL helps most where convention is least free, reducing exploitability by 26% in the most-exploitable quartile but near 0% in the least. This gradient is exactly what the low-rank theory predicts: matchups with higher exploitability have more payoff-relevant variation for the reward signal to exploit.

Projecting the CQL–BC difference onto the SVD basis confirms this directly: 94% of the distributional shift lies in the payoff null space, with only 6% in the top-3 payoff-relevant

*Table 6.* Reward sensitivity: game-theoretic quantities under varying KO weight (500 matchups, 95% CI).

| $w_{\text{ko}}$ | Exploitability | BC-vs-BC $-V^*$ | Convention free? |
|---|---|---|---|
| 1.0 | 0.58 [0.56, 0.60] | $+0.02$ [$-0.01$, 0.04] | Yes |
| 3.0 | 1.41 [1.36, 1.46] | $-0.02$ [$-0.07$, 0.04] | Yes |
| 5.0 | 2.25 [2.16, 2.33] | $-0.05$ [$-0.13$, 0.04] | Yes |

dimensions. Behaviorally, CQL's largest shift is in terastallization: it doubles the tera rate from 28% to 54% (Nash: 95%) and increases offensive targeting from 50% to 65% (Nash: 69%). This is consistent with §5—tera is the one dimension where convention is costly, so reward optimization correctly shifts probability mass there. But even this shift captures only half the Nash tera rate, reflecting CQL's conservative penalty anchoring it to the data distribution.

The key insight: the two policies are meaningfully different (TV(CQL, BC) $= 0.60$), but 94% of that movement is payoff-irrelevant. Convention persists not because imitation is the wrong objective, but because the game's low-rank structure leaves little room for *any* objective to find improvement.

## 7. Ablations and Additional Measurements

**Autoregressive BC.** An autoregressive variant with $P(a_2 \mid a_1)$ yields identical exploitability ($1.41 \rightarrow 1.41$). Mutual information between leads' actions is 0.018 bits (effectively independent), validating the factored assumption.

**Improved dynamics.** The dyn_002 model ($d_{\text{action}} = 64$, lower validation loss) finds *more* exploitability ($1.41 \rightarrow 1.52$): our base estimate is a lower bound.

**QRE rejection.** Quantal response equilibrium (McKelvey & Palfrey, 1995) fits poorly: 92% of matchups have best-fit $\lambda \leq 0.01$ (essentially uniform), with TV(QRE, BC) $= 0.68$. Experts are far more structured than "noisy Nash."

**Reward sensitivity.** Our payoff weights ($w_{\text{hp}} = 1$, $w_{\text{ko}} = 3$, $w_{\text{field}} = 0.5$) are domain-informed heuristics. We recompute all game-theoretic quantities under $w_{\text{ko}} \in \{1, 3, 5\}$ with other weights fixed (Table 6). Exploitability scales with the weight, but the cross-play gap remains indistinguishable from zero at all settings.

**Payoff-weighted TV.** Reweighting TV distance by each action's contribution to payoff variance collapses the distance from 0.99 to 0.019—a $50\times$ reduction confirming that BC–Nash divergence is concentrated in payoff-irrelevant dimensions.

**Smoothed best-response dynamics.** Smoothed BR initialized at BC converges from exploitability 2.81 to below 0.50 in 19 iterations, reaching 0.14 by iteration 500 (Figure 3).

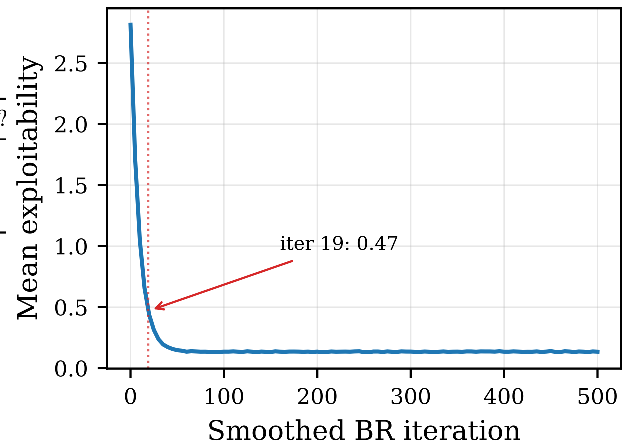


*Figure 3.* Smoothed best-response convergence from BC initialization. Exploitability decreases from 2.81 to below 0.50 in $\sim$19 iterations, reaching 0.14 at iteration 500.

**Mixture interpolation.** Blending BC toward Nash via $p_\alpha = (1-\alpha)\, p_{\text{BC}} + \alpha\, p^*$ yields a nearly perfect linear decay in exploitability: exploit$(\alpha) \approx 1.41 - 1.41\alpha$ ($R^2 = 0.9995$). There is no phase transition—no threshold $\alpha$ at which exploitability suddenly drops—consistent with 93% of the BC–Nash gap lying in the payoff null space. The geometry is flat: every step toward Nash buys exactly the same marginal improvement.

## 8. Discussion

**Most strategic variation is payoff-irrelevant.** Convention costs 1.41 reward units against an optimizer but 0.00 against another conventional player. The game's low-rank structure (effective rank 3 out of $\sim$122 actions) means 93% of strategic variation lies in the payoff null space—you can be wildly "wrong" in 119 out of 122 dimensions and it does not matter. Switching from imitation to reward optimization (CQL) narrows exploitability by 17% but does not close the gap. The central finding is not that experts are clever, but that the game is forgiving.

**Convention at every stage.** Our companion project TurnZero (Brennan-Almaraz, 2026) found that expert team selection during the pre-battle preview phase also reflects convention rather than strategic optimization (opponent contributes $-1.3$pp to prediction accuracy). Turn-1 actions exhibit the same pattern. The game's low-rank structure—which we attribute to the dominance of a few key factors (type advantage, raw power, speed tier) over hundreds of nominal action combinations—makes convention free at both stages.

**The tera exception.** Terastallization accounts for 30% of exploitability despite being a single binary decision. Unlike move selection, which lives in the payoff null space, tera is irreversible with multi-turn consequences—properties that place it outside the low-rank structure. The tension between myopic and far-sighted optimality is the main limitation of our turn-1 framework.

**Beyond Pokémon.** Two conditions—action redundancy making convention free, irreversibility making it costly—yield a testable prediction for other domains. Where the effective rank-to-size ratio of payoff matrices is small, convention should be free. Where irreversible resource commitments dominate (poker all-ins, military reserve deployment), convention should be costly.

**Limitations.** (a) Turn-1 only: multi-turn consequences, especially for tera, are not captured. (b) Dynamics model noise (SNR 3.4:1) is the primary methodological constraint. (c) The cross-play gap averages to zero but shows heterogeneity (std 0.65 per matchup); this cancellation is empirical, not a mathematical identity. (d) Regulation G specific; the low-rank hypothesis is testable in other formats. (e) Later turns with switching present higher effective rank, an open question.

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